

1.4 Materials

This topic covers density, flow of liquids, Hooke's law, the Young modulus and elastic strain energy.

This topic should be studied using a variety of applications, for example making and testing food, engineering materials, spare-part surgery for joint replacement.

This unit includes many opportunities for developing experimental skills and techniques by carrying out more than just the core practical experiments.

Candidates will be assessed on their ability to:

23	be able to use the equation density $\rho = \frac{m}{V}$
24	understand how to use the relationship upthrust = weight of fluid displaced
25	a be able to use the equation for viscous drag (Stokes' Law), $F = 6\pi\eta rv$. b understand that this equation applies only to small spherical objects moving at low speeds with <i>laminar flow</i> (or in the absence of <i>turbulent flow</i>) and that viscosity is temperature dependent
26	CORE PRACTICAL 2: Use a falling-ball method to determine the viscosity of a liquid
27	be able to use the Hooke's law equation, $\Delta F = k\Delta x$, where k is the stiffness of the object
28	understand how to use the relationships • (tensile or compressive) stress = force/cross-sectional area • (tensile or compressive) strain = change in length/original length <i>Young modulus = stress/strain.</i>
29	a be able to draw and interpret force-extension and force-compression graphs b understand the terms limit of proportionality, elastic limit, yield point, elastic deformation and plastic deformation and be able to apply them to these graphs
30	be able to draw and interpret tensile or compressive stress-strain graphs, and understand the term <i>breaking stress</i>
31	CORE PRACTICAL 3: Determine the Young modulus of a material
32	be able to calculate the elastic strain energy E_{el} in a deformed material sample, using the equation $\Delta E_{el} = \frac{1}{2} F\Delta x$, and from the area under the force-extension graph <i>The estimation of area and hence energy change for both linear and non-linear force-extension graphs is expected.</i>